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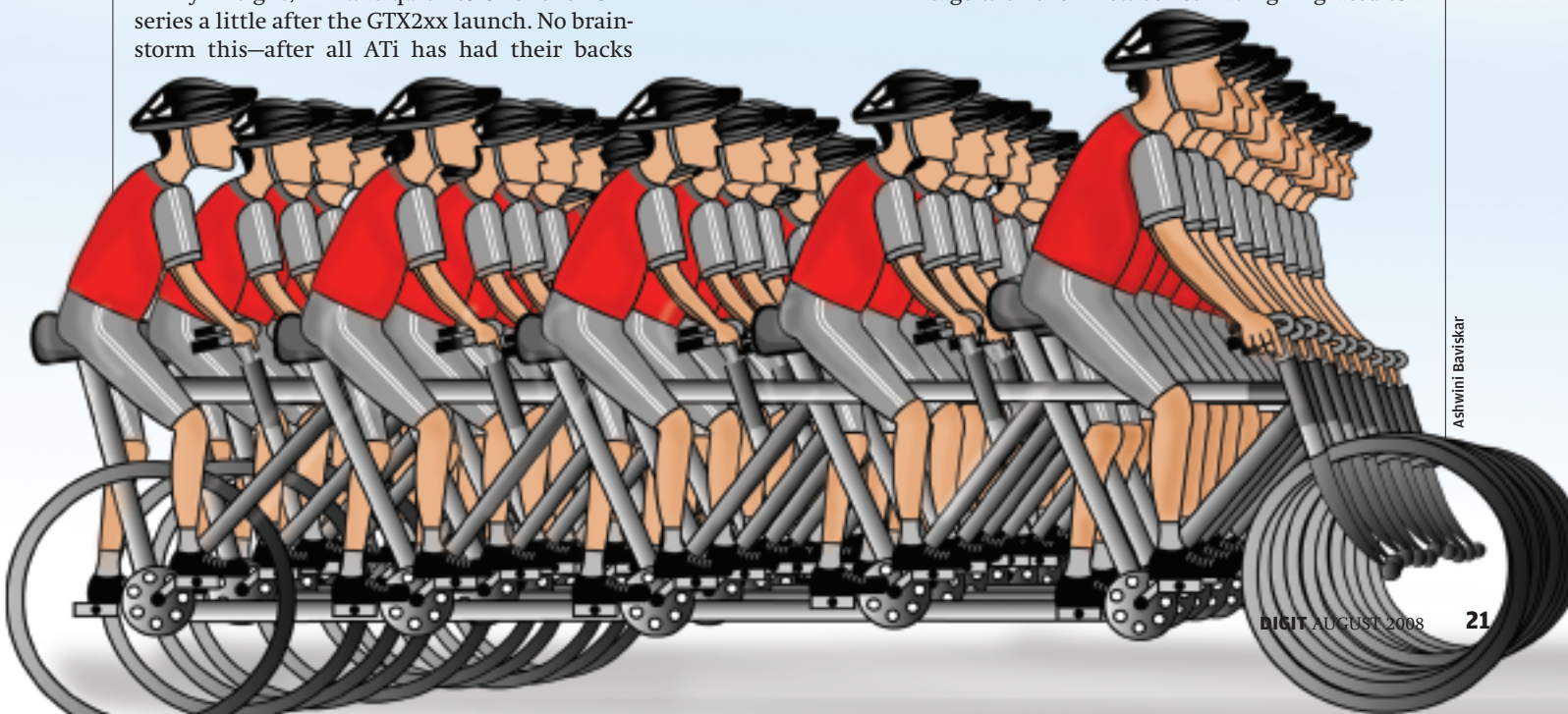
Michael Browne

Not long ago, both ATI and NVIDIA claimed to be a little more environment friendly (read green) than they had been previously, in terms of power consumption of their latest 3D accelerators. The top priority, as we know (and conflicting with anything green), is a GPU that is more powerful so that games can run faster and look better. Performance and efficiency are two binaries whose existence together is extremely rare. Looking at NVIDIA's GTX2xx architecture last month, (July 2008), we see something more powerful than their earlier G80/G92, and therefore more power hungry. We rejoice about the former, after all a boost of 30 to 75 per cent is no small matter. Complain as we may about the latter issue, there is very little power saving one can expect from a geometric increase in transistor count (from 754 million to some 1.4 billion). Power issues aside and visual nirvana firmly in sight, ATI was quick to offer the 48xx series a little after the GTX2xx launch. No brain-storm this—after all ATI has had their backs

Can ATI's new architecture do what the HD2xxx and HD3xxx families of GPUs couldn't?

solidly planted against a wall since November 2006 and the G80 (GeForce 8800GTX), which was one architecture that they (ATI) had absolutely no retort to. Their floundering position in the GPU market with failures like the HD2900XT and pseudo failures like the HD3870, while 8800GTX's sold like hot cakes is ample testament to this. ATI did have a fan following in the mid-range segment, but NVIDIA completely dominated the flagship segment, and the performance war gave way to a reign of peace with the mighty G80 trampling ATI's offerings beneath its 128 shader units a.k.a. stream processors.

With multithreading in mind, ATI's new Radeon HD4870 and HD4850 feature a staggering 800 SPs each, unheard of in any previous GPU. There're also plans to launch an X2 version, which is simply a larger PCB which houses two HD4870 GPUs therefore taking the SP count to a colossal 1,600 SPs. In terms of fabrication processes, ATI has a slight advantage with their new series having migrated to a



Ashwini Baviskar